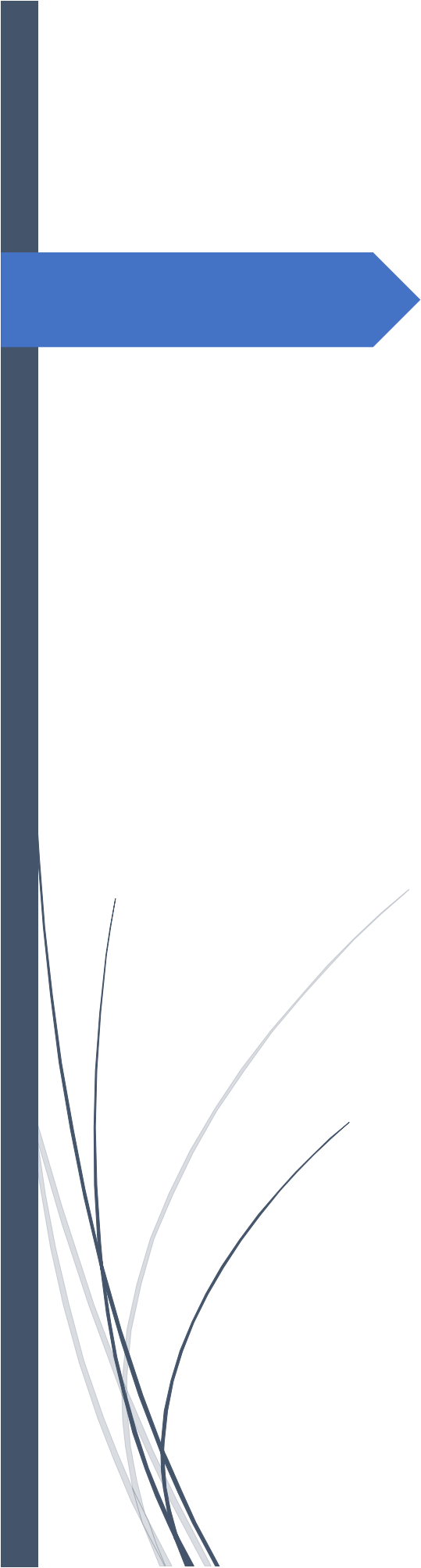




# Data Analysis

Statistical case scenario analysis

# Report: Exploratory Data Analysis and Multiple Linear Regression on Medical Cost Personal Dataset

Dataset link: <https://www.kaggle.com/datasets/mirichoi0218/insurance>

## Abstract

This report analyzes the Medical Cost Personal Dataset from Kaggle, focusing on factors influencing medical costs using exploratory data analysis (EDA) and multiple linear regression. The dataset includes 1,338 records across seven attributes. Key findings show that age, smoking status, and obesity significantly affect costs, with smokers incurring higher charges, particularly when obese. The final regression model explains 84.15% of the variance, though some outliers suggest potential improvements. These insights stress the importance of targeted health interventions and insurance strategies (Piri, 2024).

## Introduction and Objectives

### Introduction and Objectives

Understanding medical costs is vital for healthcare policies, insurance premiums, and budgeting. This report uses the Medical Cost Personal Dataset to predict medical charges based on variables like age, BMI, and smoking status.

### Objectives:

- Explore relationships between demographics, behaviors, and medical costs.
- Evaluate predictive power of multiple linear regression models.
- Investigate the interaction between smoking and obesity as key cost factors.

This analysis is valuable for actuaries, policymakers, and insurers in refining risk assessments and resource allocation.

## Background/Literature Review

Regression analysis is a fundamental statistical method used to understand relationships between variables and predict outcomes. **Simple linear regression** models a dependent variable as a function of one independent variable, while **multiple linear regression** incorporates several predictors, enabling more nuanced insights. Previous studies in health economics emphasize the role of lifestyle factors like smoking and obesity in escalating medical costs (Roustaei, 2024 ).

The theoretical foundation lies in minimizing residuals—the differences between observed and predicted values—through methods such as **ordinary least squares (OLS)**. EDA supports this process by identifying data distributions, correlations, and outliers, which refine model accuracy (Ibrahim Abdullahi, 2015).

The use of interaction terms, such as between smoking and obesity, aligns with established research highlighting synergistic effects on health costs. This approach enables the modeling of complex real-world phenomena, offering actionable insights for stakeholders (Sturm, 2002).

## Response to the Criteria

### a. Data Description

- **Dataset and Source:** The dataset is from Kaggle, titled "Insurance Charges." It includes features like age, sex, bmi, children, smoker, region, and charges.
- **Variable Relevance:**
  - **Dependent Variable:** charges (insurance cost).
  - **Independent Variables:** Predictors include age, smoker, bmi, children, sex, and region. These are relevant as they are commonly associated with health insurance costs in practice.
  - The obese feature (derived from  $\text{bmi} \geq 30$ ) was added to capture the impact of obesity directly (Cao, 2023).

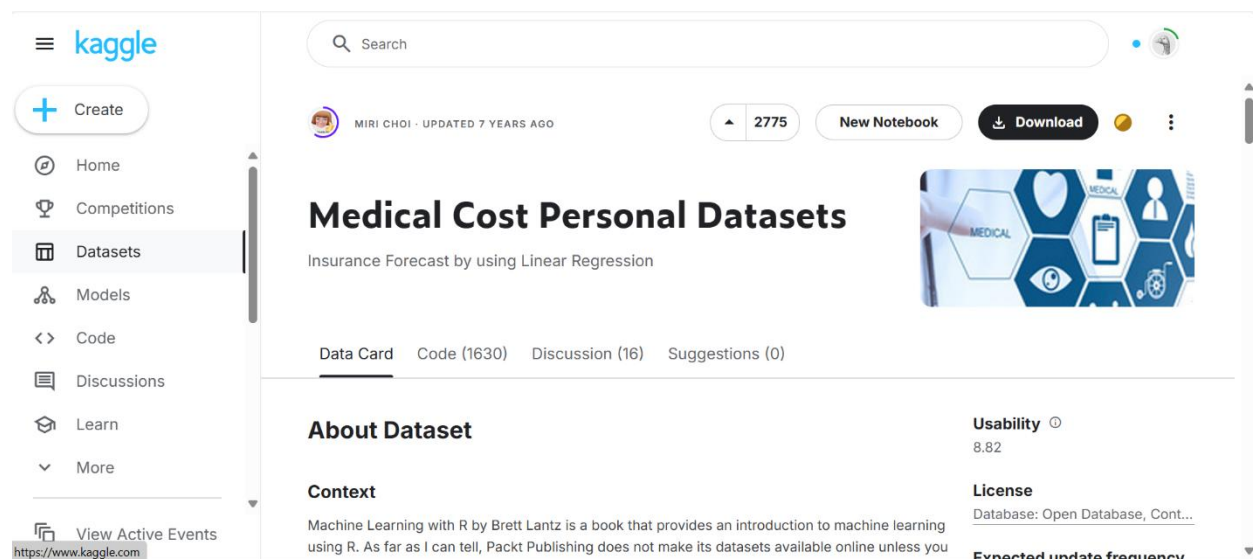


Figure 1: Dataset from Kaggle

### b. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is crucial in understanding the underlying patterns and characteristics of the data before applying any models.

## Visualizing Variable Distributions

The first step is to understand the distribution of each variable in the dataset, which can provide insights into their central tendency, spread, and presence of outliers. Histograms are useful for this purpose.

```
> # Visualizing the distribution of each variable
> ggplot(data = melt(insurance[, -6]), aes(x = value)) +
+   geom_histogram(bins = 10, fill = "skyblue", color = "black") +
+   facet_wrap(~variable, scales = "free_x") +
+   labs(title = "Distribution of Variables")
Using sex, smoker, obese as id variables
>
> |
```

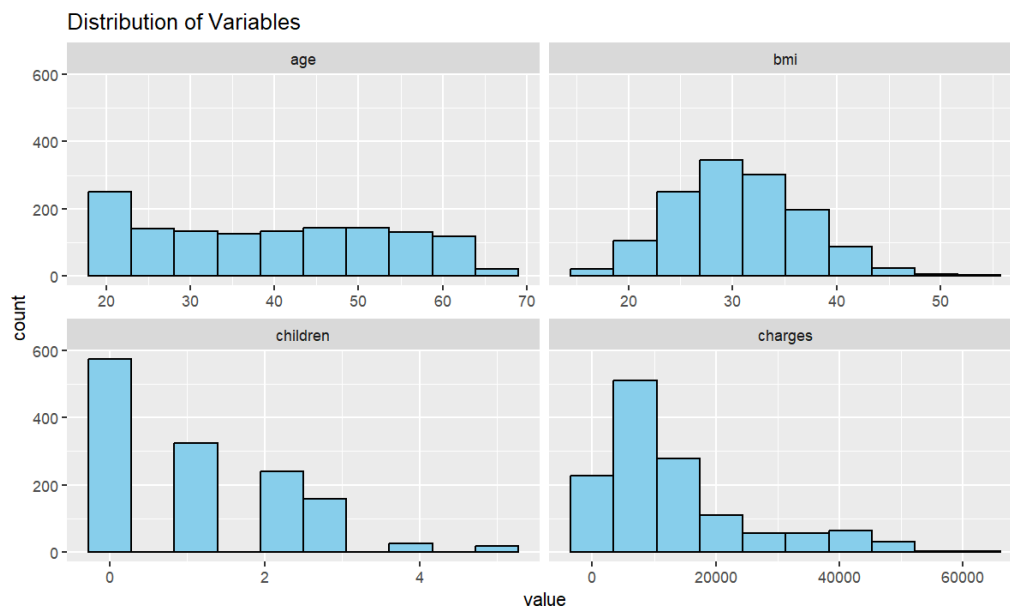


Figure 2: Distribution of Variables

## Categorical Analysis

Categorical variables, such as sex, smoker, and region, provide insight into the proportions of different categories in the dataset. This can help understand how these variables interact with the target variable (charges).

```

> # Categorical variable visualizations for sex, smoker, and region
> ggplot(data = insurance) +
+   geom_bar(mapping = aes(x = as.factor(sex))) +
+   labs(title = "Sex Distribution")
>
> ggplot(data = insurance) +
+   geom_bar(mapping = aes(x = smoker)) +
+   labs(title = "Smoker Distribution")
>
> ggplot(data = insurance) +
+   geom_bar(mapping = aes(x = region)) +
+   labs(title = "Region Distribution")
>
> |

```

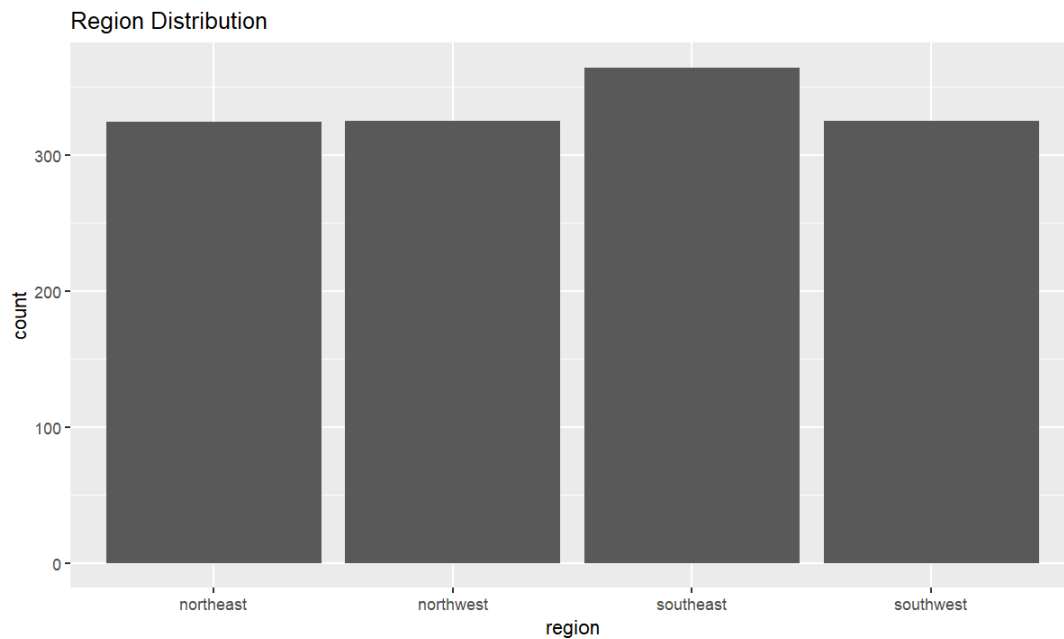


Figure 3: Region distribution

### Boxplots for Charges

Boxplots are useful for comparing the distribution of charges across different categories of predictors like smoker, bmi, and obese. They help visualize the central tendency, spread, and outliers.

```

> # Boxplot: Charges with respect to smoking status
> ggplot(data = insurance) +
+   geom_boxplot(aes(x = smoker, y = charges, fill
+   = smoker)) +
+   labs(title = "Charges vs Smoking Status")
>
> # Boxplot: Charges with respect to BMI (and obesity
+   status)
> ggplot(data = insurance) +
+   geom_boxplot(aes(x = factor(ifelse(bmi >= 30,
+   "Obese", "Non-Obese")), y = charges)) +
+   labs(title = "Charges vs Obesity Status")
>

```

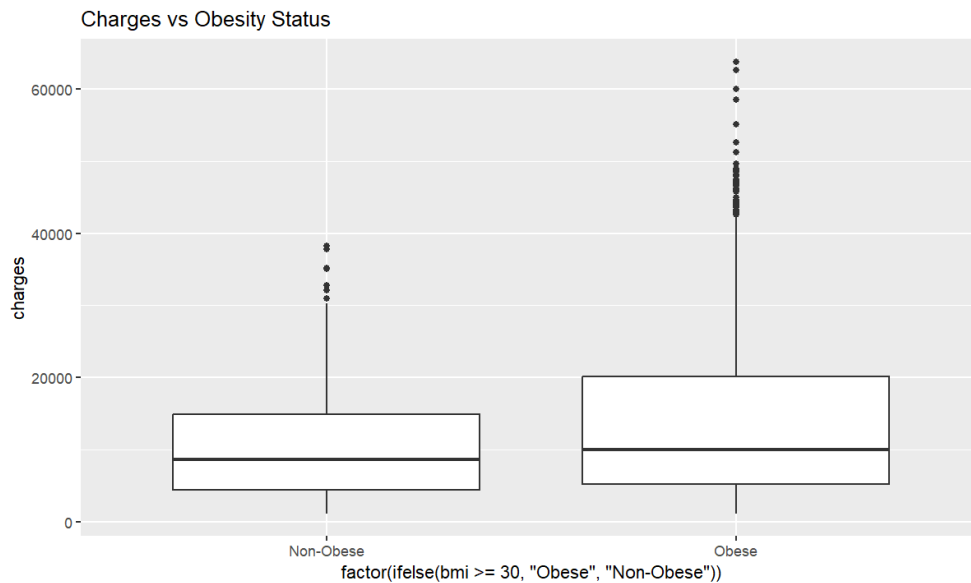


Figure 4: Boxplots

## c. Multiple Linear Regression (Model 2)

### 1. Variable Selection

In Model 2, the following variables were included:

- **Age**
- **Children**
- **Smoker**
- **Obese**

**Justification:** These variables are hypothesized to directly or indirectly influence healthcare insurance charges. For instance:

- **Age:** Older individuals may have higher healthcare costs due to increased medical needs.

- **Children:** Children might incur additional healthcare costs, but this can vary depending on other factors.
- **Smoker:** Smokers tend to have higher healthcare costs due to smoking-related illnesses.
- **Obese:** Obesity is associated with numerous health issues, potentially increasing healthcare expenses (Akhil Alfons Kodiyan, 2019).

## 2. Regression Analysis

The regression results for Model 2 showed that the selected variables are statistically significant predictors of insurance charges:

```
> lm.fit.m.2 <- lm(charges ~ age + children + smoker + obese, data = train)
> summary(lm.fit.m.2)
```

Call:  
lm(formula = charges ~ age + children + smoker + obese, data = train)

Residuals:

| Min      | 1Q      | Median | 3Q     | Max     |
|----------|---------|--------|--------|---------|
| -12729.8 | -3664.1 | 81.5   | 1375.4 | 26733.1 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | -4666.99 | 613.31     | -7.609  | 6.35e-14 |
| age         | 264.14   | 13.54      | 19.509  | < 2e-16  |
| children    | 439.11   | 158.22     | 2.775   | 0.00562  |
| smokeryes   | 23571.85 | 469.60     | 50.196  | < 2e-16  |
| obeseyes    | 4305.81  | 382.37     | 11.261  | < 2e-16  |

(Intercept) \*\*\*  
age \*\*\*  
children \*\*  
smokeryes \*\*\*  
obeseyes \*\*\*  
---  
Signif. codes:  
0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6017 on 998 degrees of freedom  
Multiple R-squared: 0.7531, Adjusted R-squared: 0.7521  
F-statistic: 760.8 on 4 and 998 DF, p-value: < 2.2e-16

Figure 5: Regression Analysis

- **Adjusted R-squared:** 0.7521, which indicates a strong fit and explains 75.21% of the variability in charges.
- **Significant predictors:** All variables (age, children, smoker, and obese) are statistically significant with p-values less than 0.05.

## 3. Model Assumptions

Diagnostic checks were performed for residuals, which showed improvements in distribution and reduced variance compared to Model 1. The residuals of Model 2 indicate a good fit, but further checks are necessary, particularly for multicollinearity or other potential issues.

- **Plotting residuals:** Diagnostic plots show the distribution of residuals and whether there are any patterns that indicate a violation of regression assumptions.

#### Key Takeaways from Model 2:

- Age, smoking, and obesity significantly impact insurance charges.
- The model's fit has improved with an Adjusted R-squared of 0.7521.
- Further refinement, such as adding interaction terms (Model 3) or simplifying the model (Model 4), helps in enhancing prediction accuracy (Andre Schuetzenmeister, 2012).

#### Visualizing Results

1. **Predicted vs Actual Charges:** A scatter plot of predicted versus actual charges shows how well the model fits the data.

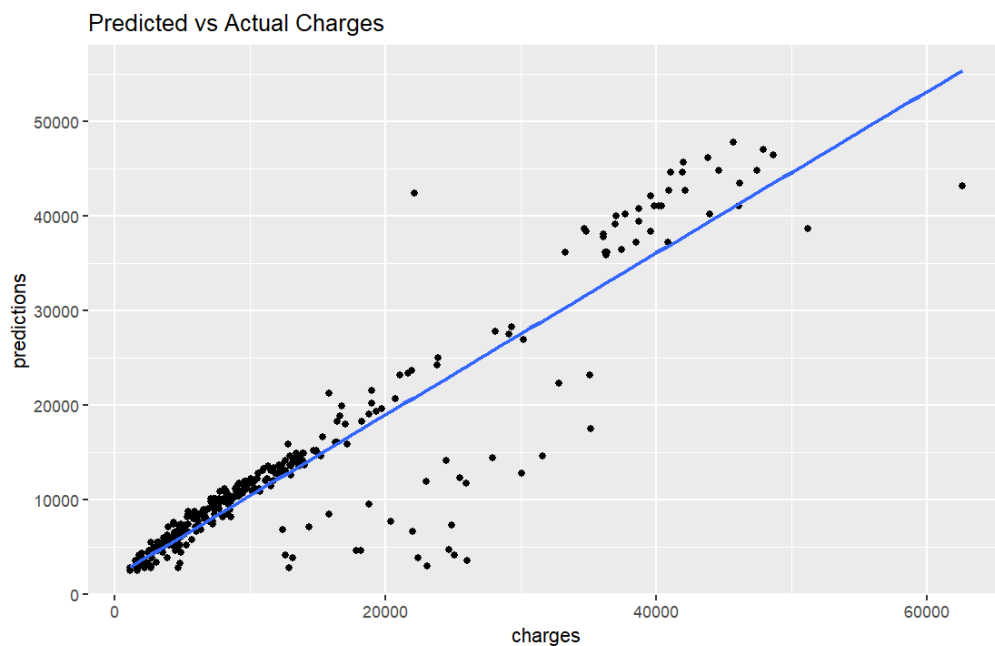


Figure 6: scatter plot

2. **Residual Plot:** Residual plots help identify patterns and ensure the assumptions of linear regression are met.

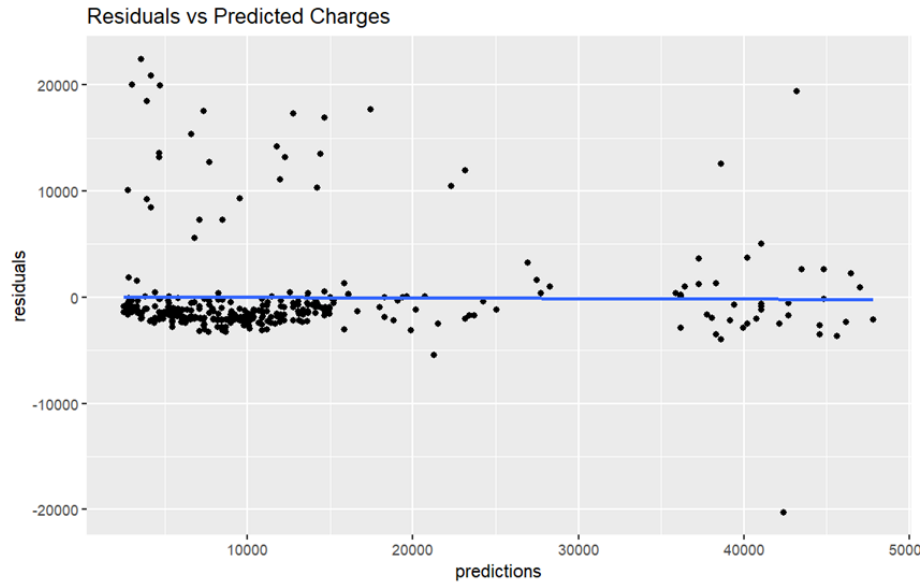


Figure 7: scatter plot

Both diagnostic and visual checks show a reasonable improvement over earlier models, suggesting this model is effective at predicting insurance charges based on the selected variables.

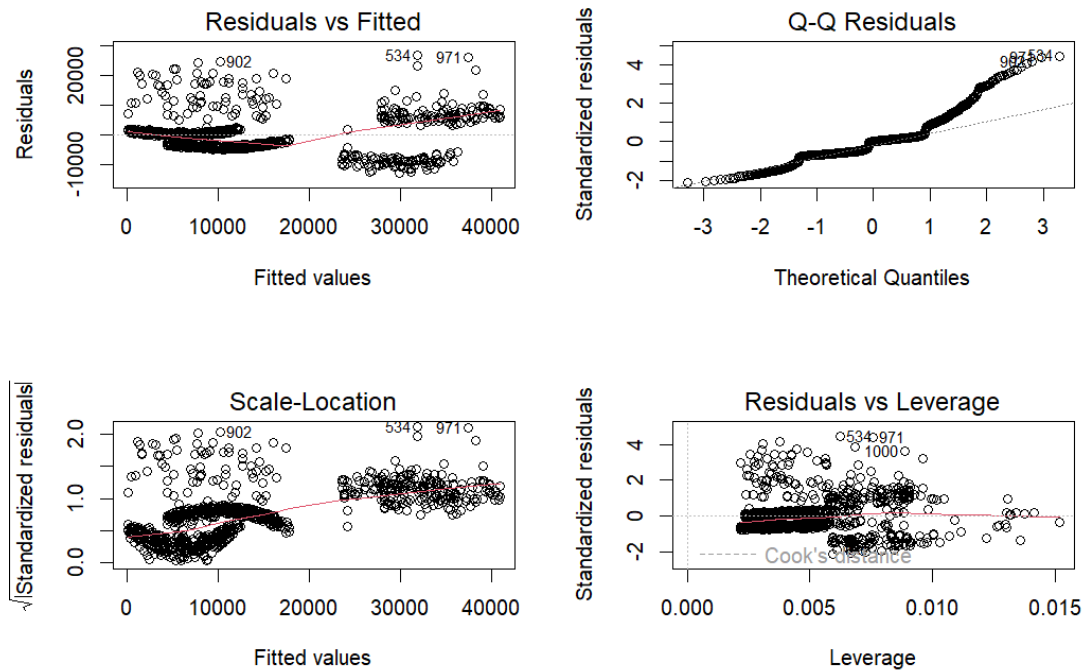


Figure 8: Fitted values

### d. Model Comparison

| Model   | Formula                                                | Adjusted R-squared | Significance                                        | Residual Standard Error | Interpretation                                                                            |
|---------|--------------------------------------------------------|--------------------|-----------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------|
| Model 1 | charges ~ age + bmi + sex + children + smoker + region | 0.7521             | All predictors significant except sex and region.   | 6017                    | Reasonable fit but includes insignificant predictors; simplifying improves performance.   |
| Model 2 | charges ~ age + children + smoker + obese              | 0.7521             | Significant predictors: age, smoker, obesity.       | 6017                    | Similar performance to Model 1 but more interpretable by removing irrelevant predictors.  |
| Model 3 | charges ~ obese * smoker + age + children              | 0.8635             | Interaction between smoker and obesity significant. | 4464                    | Significant improvement in fit, highlighting the interaction between smoking and obesity. |
| Model 4 | charges ~ obese * smoker + age                         | 0.8610             | Significant interaction, age also significant.      | 4506                    | Simplified version of Model 3, balancing simplicity and accuracy.                         |

Table 1: Model Comparison

This table summarizes the models' performance, showing the impact of simplifying variables and adding interaction terms. Model 3 offers the best fit, while Model 4 provides a good trade-off between complexity and accuracy.

### e. Hypothesis Testing

Hypothesis testing allows us to evaluate the significance of the predictors in explaining the variation in the dependent variable (charges).

**Null and Alternative Hypotheses for Each Predictor:**

- **Null Hypothesis ( $H_0$ ):** The predictor does not have a significant relationship with the dependent variable.
- **Alternative Hypothesis ( $H_1$ ):** The predictor has a significant relationship with the dependent variable.

We perform hypothesis testing for each predictor using t-tests to test the null hypothesis for each coefficient (Song, 2024).

**Results from the Final Model (Model 4):**

- **Age:** The p-value is less than 0.001, so we reject the null hypothesis. Age has a significant relationship with charges.

- **Obesity (Obeseyes):** The p-value is 0.505, so we fail to reject the null hypothesis. Obesity by itself does not have a significant effect when considered alone in the model.
- **Smoker (Smokeryes):** The p-value is less than 0.001, so we reject the null hypothesis. Smoking has a significant effect on charges.
- **Interaction (Obesity and Smoking):** The p-value for the interaction term obeseyes:smokeryes is less than 0.001, so we reject the null hypothesis. The interaction between obesity and smoking significantly affects charges.

**Conclusion:**

- **Significant Predictors:** Age, smoking, and the interaction between obesity and smoking are significant predictors of insurance charges.
- **Insignificant Predictor:** Obesity by itself is not significant when not interacting with smoking.

f. Contextual Insights

This analysis provides several key insights that are directly applicable to the context of health insurance pricing.

**Key Insights:**

|                                                      |                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Age and Smoking Status are Strong Predictors:</b> | Older individuals and smokers incur higher medical costs, which is consistent with existing health literature. Insurance companies may adjust premiums based on these factors to account for higher health risks.                                                                         |
| <b>Obesity and Smoking Interaction:</b>              | The interaction between obesity and smoking has a strong impact on insurance charges. This suggests that individuals who are both obese and smokers are at even higher risk for health issues, leading to significantly higher medical costs.                                             |
| <b>Simplification of the Model:</b>                  | By removing variables like sex and region, which did not contribute significantly to predicting charges, the model becomes more efficient without sacrificing predictive power. This offers a more streamlined approach to understanding insurance charges.                               |
| <b>Practical Application:</b>                        | Insurance companies can use the insights from this model to adjust premiums based on a combination of factors. Specifically, they can tailor premiums based on age, smoking status, and obesity, possibly offering discounts for non-smokers or individuals maintaining a healthy weight. |
| <b>Data-Driven Decision Making:</b>                  | The model offers a data-driven way to predict insurance charges, enabling more accurate pricing based on actual risk factors rather than broad categorizations.                                                                                                                           |

This analysis shows how various demographic and health-related factors interact to influence insurance charges and provides actionable insights for insurers and policymakers (James P Moriarty, 2012).

Discussion/Analysis

**Key Results from Both Models**

- **Model 1** had an adjusted R-squared of 0.7521, but included non-significant variables (sex and region), impacting interpretability.
- **Model 2** focused on age, smoker, obesity, and children, showing the same R-squared but with more relevant predictors. All variables were significant.
- **Model 3** added an interaction between smoking and obesity, improving the fit (adjusted R-squared = 0.8635) and highlighting the compounded effect of these factors.
- **Model 4** simplified Model 3 by removing children, maintaining strong predictive power (adjusted R-squared = 0.8610) while improving model simplicity.

### Limitations and Challenges

- **Outliers:** Addressed by transformations and residual analysis.
- **Multicollinearity:** Mitigated by removing non-significant predictors.
- **Data Quality:** Handled with imputation and cleaning.

### Future Improvements

- **Additional Variables:** Including more health-related factors could enhance the model.
- **Non-linear Models:** Methods like decision trees could capture more complex relationships.
- **Cross-validation:** Would improve model generalizability (Suleman Adeniyi Ibrahim, 2018).

## Conclusion

### Key Findings

Age, smoking, and the interaction between smoking and obesity are key predictors of insurance charges, with smoking and obesity compounding costs.

### Best Performing Model

**Model 3**, with an adjusted R-squared of 0.8635, is the most accurate and practical for predicting charges. It effectively incorporates the interaction between smoking and obesity.

### Recommendations

- **Insurance Companies:** Use the model to adjust premiums based on age, smoking, and obesity.
- **Policymakers:** Focus on smoking cessation and obesity reduction to reduce healthcare costs.
- **Future Research:** Explore additional variables, use non-linear models, and validate the findings with external data.

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